

# Long title that appears on the title page

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## Abstract

Shell document for REV<sub>T</sub>EX 4.

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## I. ENERGY FUNCTIONAL

$$\begin{aligned} \mathcal{E}(\mathbf{n}, \psi, \theta) = & \frac{\mathcal{H}(\psi, \mathbf{n})}{S_N(\mathbf{n})} = \frac{1}{S_N(\mathbf{n})} \left\{ \sum_{1 \leq j \leq s} N_j \{ \langle \psi_j | h_1 \psi_j \rangle + \langle \psi_{j+s} | h_1 \psi_{j+s} \rangle + \langle \psi_j \psi_{j+s} | h_2 | \psi_j \psi_{j+s} \rangle \} \right. \\ & + \sum_{1 \leq j, k \leq s} n_j^{\frac{1}{2}} n_k^{\frac{1}{2}} S_{N-1} \left( \hat{j} \hat{k} \right) e^{i(\theta_j - \theta_k)} \langle \psi_j \psi_{j+s} | h_2 | \psi_k \psi_{k+s} \rangle + \sum_{1 \leq i, j \leq s} n_i n_j S_{N-2} \left( \hat{j} \hat{k} \right) \langle \psi_j \psi_k | | \psi_j \psi_k \rangle \} \end{aligned} \quad (1)$$

where  $\psi$  are the NGSO's of  $|g^N\rangle \in [D^1]_N$ ,  $\mathbf{0} \leq \mathbf{n} \leq \mathbf{I}, Tr \{ \mathbf{n} \} = 1, \mathbf{0} \leq \theta < 2\pi$ .

## II. EXAMPLE

$$\begin{aligned} |g^2\rangle = & (c_1 |\psi_1 \psi_4\rangle + c_2 |\psi_2 \psi_5\rangle + c_3 |\psi_3 \psi_6\rangle)^2 = c_1 c_2 |\psi_1 \psi_2 \psi_3 \psi_4\rangle + c_1 c_3 |\psi_1 \psi_2 \psi_5 \psi_6\rangle \\ & + c_2 c_3 |\psi_3 \psi_4 \psi_5 \psi_6\rangle \end{aligned} \quad (2)$$

The following geminals have the same FORDO

$$\begin{aligned} D^2(g^N(\theta_1)) = & (S_N)^{-\frac{1}{2}} \left\{ \sum_{1 \leq j \leq s} N_j |\psi_j \psi_{j+s}\rangle \langle \psi_j \psi_{j+s}| + \sum_{1 \leq j, k \leq s} n_j^{\frac{1}{2}} n_k^{\frac{1}{2}} e^{i(\theta_{1j} - \theta_{1k})} S_{N-1} \left( \hat{j} \hat{k} \right) |\psi_j \psi_{j+s}\rangle \langle \psi_k \psi_{k+s}| \right. \\ & \left. + \sum_{1 \leq j < k \leq s} n_j n_k S_{N-2} \left( \hat{j} \hat{k} \right) |\psi_j \psi_k\rangle \langle \psi_j \psi_k| \right\}, \end{aligned} \quad (3)$$

$$\theta_{1jk} = \theta_{1j} - \theta_{1k} = \theta_{2j} - \theta_{2k} = \theta_{2jk} \quad \forall j, k \quad (4)$$

$$|g\rangle = c_1 e^{\theta_1} |\psi_1 \psi_4\rangle + c_2 e^{\theta_2} |\psi_2 \psi_5\rangle + c_3 e^{\theta_3} |\psi_3 \psi_6\rangle \quad (5)$$

$$\theta_2 = \theta_1 + \varphi_1, \theta_3 = \theta_1 + \varphi_2 \quad (6)$$

$$F |\psi_j\rangle = \varepsilon_j |\psi_j\rangle \iff Fe^{\varphi_j} |\psi_j\rangle = \varepsilon_j e^{\varphi_j} |\psi_j\rangle \quad (7)$$

$$\varphi_j + \varphi_{j+s} = \theta_j \quad (8)$$

$$\theta_{jk} = \theta_j - \theta_k = \varphi_j + \varphi_{j+s} - \varphi_k - \varphi_{kj+s} \quad (9)$$

If  $\theta_{jk} \neq 0 \ \forall jk$  then the solutions refer to a different geminal.

$$(10)$$